

terms directly written into code. They automatically execute and enforce agreements when predefined conditions are met.

Identity verification: Blockchain can be used to securely manage digital identities and reduce the risk of identity theft.

Voting systems: Blockchain can enhance the security and transparency of voting systems, reducing the potential for fraud.

Healthcare: Patient records can be securely stored and shared among healthcare providers using blockchain, ensuring data accuracy and privacy.

Real estate: Blockchain can streamline property transactions, reducing paperwork and improving efficiency.

Financial services: Blockchain has the potential to revolutionise traditional financial systems by enabling faster, more secure, and cheaper cross-border transactions.

Whether you're considering launching a decentralised application (DApp), managing digital assets, or exploring supply chain solutions, selecting the right blockchain platform is a pivotal decision.

Public blockchains and private blockchains

A fundamental distinction in the world of blockchain is the division between public blockchains and private blockchains.

The choice between public and private blockchains depends on the specific needs of your project or organisation. Public blockchains excel in scenarios where decentralisation, openness, and transparency are paramount, while private blockchains offer control and privacy. The emergence of hybrid blockchains provides even more flexibility.

We'll delve into the nuances of these two blockchain variants and explore their use cases, benefits, and trade-offs.

Public blockchains: The power of permissionlessness

Permissionless public blockchains are perhaps the most iconic face of blockchain technology. They epitomise decentralisation and openness. Here's what characterises them.

Accessibility for all

Permissionless public blockchains are open to anyone, anywhere. Participation requires no prior approval or vetting. This inclusive nature enables global participation, fostering a truly decentralised ecosystem.

Transparency and trustlessness

Transactions on permissionless public blockchains are transparent and immutable. They're recorded on a public ledger that anyone can access and verify. Trust is established not through intermediaries but through cryptographic algorithms and consensus mechanisms.

Prime examples

Two of the most renowned examples of permissionless public blockchains are Bitcoin and Ethereum. Bitcoin introduced the world to the concept of a decentralised digital currency, and Ethereum pioneered smart contracts, enabling programmable, self-executing agreements.

Use cases

Permissionless public blockchains are ideal for applications where trust, censorship resistance, and decentralisation are paramount. They are often used for digital currencies, DApps (decentralised applications), and scenarios where transparency is critical, like voting systems and public records.

Private blockchains: Control and privacy

Permissioned private blockchains, on the other hand, introduce a layer of control and privacy into the blockchain landscape. Here's what defines them.

Controlled access

Access to a permissioned private blockchain is restricted. Participants are known and typically vetted, ensuring a degree of trust among network members. These blockchains often cater to consortiums, businesses, or organisations.

Enhanced privacy

Privacy is a key feature of permissioned private blockchains. Unlike their public counterparts, transactions on these networks are not necessarily visible to all participants. They provide a level of confidentiality that can be crucial in enterprise settings.

Use cases

Permissioned private blockchains find applications in contexts where privacy, regulatory compliance, and control are vital. Industries like finance, healthcare, and supply chain management use them to streamline processes, secure data, and meet legal requirements.

Hybrid blockchains: The best of both worlds

In some cases, a middle ground is sought through hybrid blockchains. These blockchains combine features of both public and private blockchains. They offer selective transparency and access control, making them adaptable to various use cases. Hybrid blockchains are an excellent solution when there's a need for shared data among multiple parties while maintaining a degree of confidentiality.

Blockchain platforms

Open source blockchain projects provide transparency, flexibility, and often have active developer communities. There are several available. Table 1 lists some notable open source blockchain platforms and projects.

These are just a few examples of open source blockchain projects, and there are many more in various stages of development. When choosing an open source blockchain platform, consider